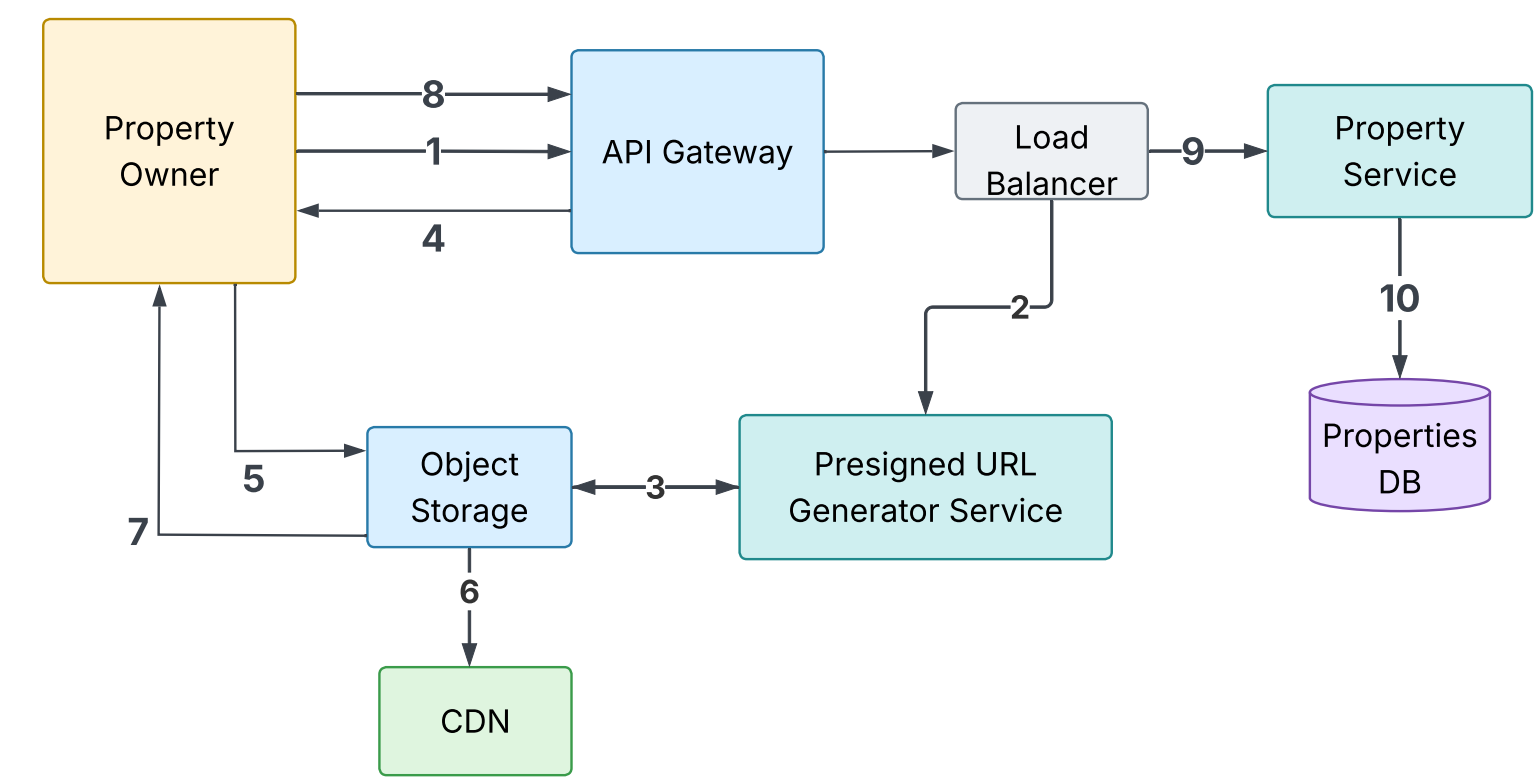


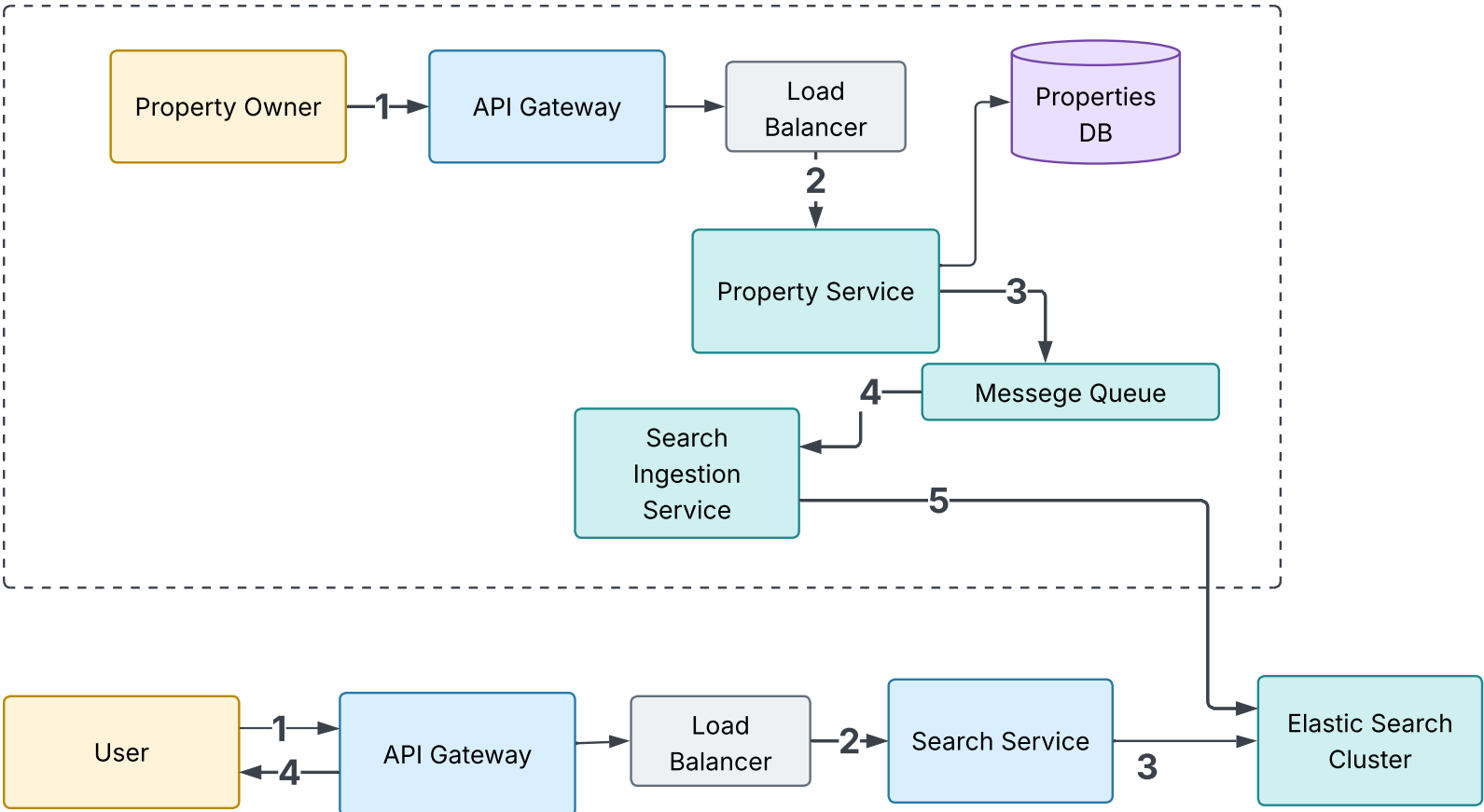
# Airbnb Architecture Design

## 1. Add Property



- 1 - The owner request to add property goes through API gateway.
- 2 - Request is sent to a presigned URL Generator Service Via Load Balancer.
- 3 - It asks object storage to generate the presigned URLs for each image
- 4 - It then send the URLs back to the user.
- 5 - User then uploads the images directly to the Object Storage via the URLs.
- 6 - Object Storage pushes the images to the CDN.
- 7 - URLs of the uploaded images are returned to the user.
- 8 - User clicks send request (with images) to the API gateway, finally list property by clicking publish.
- 9 - The API gateway Requests Property Service via load balancer.
- 10 - Property Services stores the request data into the Properties DB.

## 2. Search Property



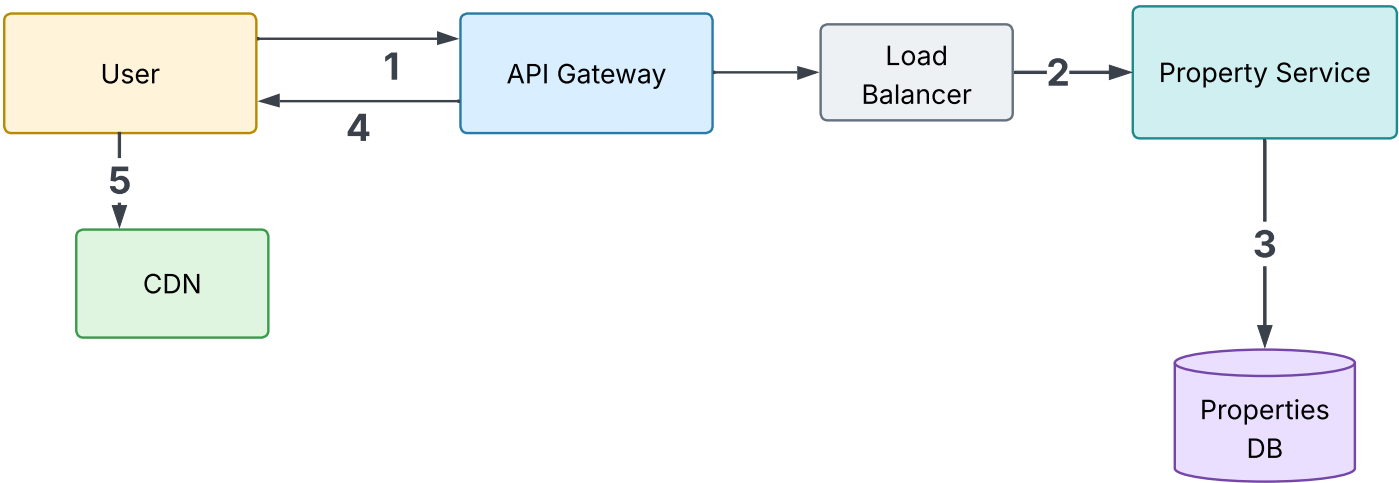
### User side

- 1 - The user request to search property goes through API gateway.
- 2 - The API gateway Requests Search Service via load balancer.
- 3 - The search service uses elastic search cluster for the user search.
- 4 - Search Results are returned back to the user.

### Property Owner side

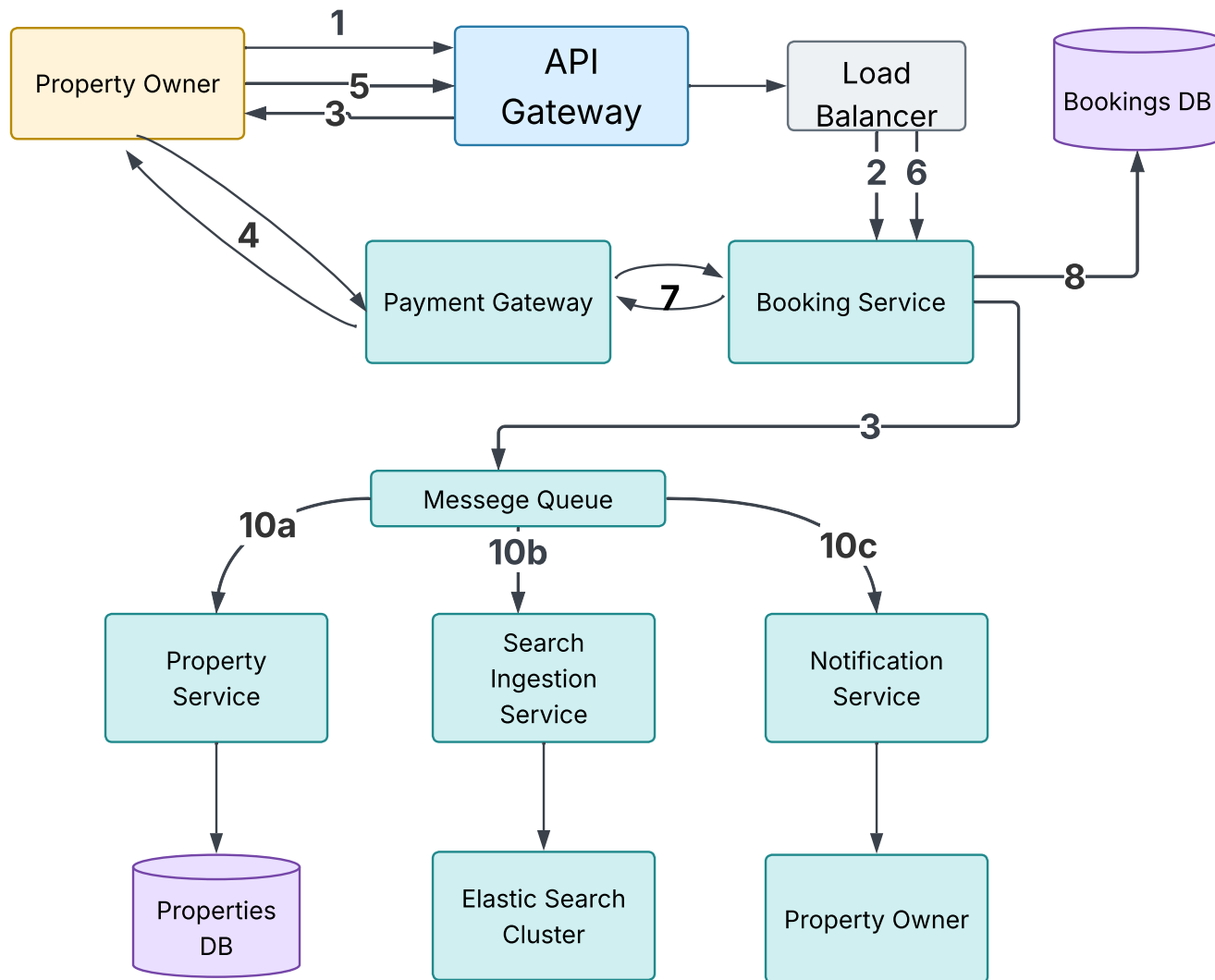
- 1 - The owner request to add property goes through API gateway.
- 2 - The API gateway Requests Search Service via load balancer.
- 4 - The Property Service creates an event with all the property details in a message queue.
- 4 - Search Ingestion Service listens for this event in the message queue.
- 5 - The Search Ingestion Service Delivers the data to the Elastic Search Cluster.

## 3. View Property



- 1 - The user request to view property goes through API gateway.
- 2 - The API gateway Requests Property Service via load balancer.
- 3 - The property service fetches the property details from the properties DB.
- 4 - The property details are returned to the user.
- 5 - The images are recieved through a CDN.

## 5. Book Property



1. The user request to book property goes through API gateway.
2. The API gateway Requests Booking Service via load balancer.
3. The booking service sends a payment URL back to the user.
4. This URL takes the user to a payment gateway. After completing the payment/transaction, the Payment Gateway will provide a transaction completion token back to the user as a proof of transaction.
5. User will send this token to the API Gateway.
6. The API gateway sends the transaction token to the booking service via load balancer.
7. The Booking Service then verifies the token with the Payment Gateway to verify and approve the transaction. The payment gateway send a confirmation to the booking service to proceed.
8. After confirming, the booking service will create a booking in the bookings database.
9. It also creates an event with all booking details for a booking and send it to the message queue.

There are 3 services that listen to this message queue:

10a - Property Service would extract the check in and check out date from the Event and update those in the properties database.

10b - Search Ingestion Service would extract the check in and check out date from the Event and update the availability of the property in the search data.

10c - Notification Service will get the owners ID from the Event and use that to send notification to the Owner of the booked property.

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